

Delta Method and Method of Moments

Clément Lalanne

Notation for this lecture

Parameter and observations. The parameter is $\theta \in \Theta \subset \mathbb{R}^p$, the observations $X_1, X_2, \dots$ are i.i.d. copies of a random vector X of $\mathbb{R}^d$, and $\hat{\theta}_n$ denotes an estimator built from $X_1, \dots, X_n$.

Derivatives. For $\phi : \mathbb{R}^p \rightarrow \mathbb{R}^m$ differentiable at θ , we denote by $J\phi(\theta)$ its $m \times p$ Jacobian matrix at θ . When $m = 1$, $\nabla\phi(\theta)$ denotes the gradient column vector and $H\phi(\theta)$ the Hessian.

Model. A statistical model is $\{\mathcal{L}_\theta; \theta \in \Theta\}$, each $\mathcal{L}_\theta$ a law on $\mathbb{R}^d$. We write $\mathbb{E}_{X \sim \mathcal{L}_\theta}$, abbreviated $\mathbb{E}_\theta$, for expectations computed as if the parameter were θ , which is the only abbreviation of this lecture.

Sample quantities. We denote $\bar{X}_n = \frac{1}{n} \sum_{i \leq n} X_i$, and any sample dependent object carries its index (e.g. $\hat{\theta}_n$, $\hat{\mu}_{2,n}$, e_n).

Part I. The delta method

The question

Setup. Given a sequence of estimators $(\hat{\theta}_n)$ of a parameter θ and a mapping ϕ , it is natural to estimate $\phi(\theta)$ by $\phi(\hat{\theta}_n)$.

Question. We ask whether the asymptotic properties of $\hat{\theta}_n - \theta$ are inherited by $\phi(\hat{\theta}_n) - \phi(\theta)$.

Consistency. Here the answer is immediate. If $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta$ and ϕ is continuous at θ , then $\phi(\hat{\theta}_n) \xrightarrow{\mathbb{P}} \phi(\theta)$ by continuous mapping, since the continuity set contains the constant θ .

Rate and limit law. Let us now suppose that

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} \mathcal{N}(0, \Sigma) .$$

In contrast, nothing above says at what rate $\phi(\hat{\theta}_n) - \phi(\theta)$ goes to zero, nor what its limit law is.

The linear case

If ϕ is linear, given by an $m \times p$ matrix M , then $\phi(\hat{\theta}_n) - \phi(\theta) = M(\hat{\theta}_n - \theta)$, and we have

$$\begin{aligned}\sqrt{n}(M\hat{\theta}_n - M\theta) &= M\sqrt{n}(\hat{\theta}_n - \theta) \\ &\xrightarrow{d} M\mathcal{N}(0, \Sigma) \quad \text{by continuous mapping with } x \mapsto Mx \\ &= \mathcal{N}(0, M\Sigma M^\top) .\end{aligned}$$

The last equality follows from the characteristic function computation of lecture 2, since for $Z \sim \mathcal{N}(0, \Sigma)$ and $t \in \mathbb{R}^m$, we have $\mathbb{E}_Z[e^{it^\top MZ}] = \mathbb{E}_Z[e^{i(M^\top t)^\top Z}] = \exp(-\frac{1}{2}t^\top M\Sigma M^\top t)$.

The intuition of the delta method. A function differentiable at θ is, near θ , roughly its linearisation. Hence, since $\hat{\theta}_n$ concentrates at θ , only the behaviour of ϕ at θ should matter, and $J\phi(\theta)$ should play the role of M .

The theorem

Theorem (delta method). Let $\theta \in \mathbb{R}^p$ and let $\phi : \mathbb{R}^p \rightarrow \mathbb{R}^m$ be differentiable at θ . Let $(\hat{\theta}_n)$ be a sequence of random vectors of $\mathbb{R}^p$, X a random vector of $\mathbb{R}^p$, and (r_n) real numbers with $r_n \rightarrow \infty$. If $r_n(\hat{\theta}_n - \theta) \xrightarrow{d} X$, then

$$r_n \left(\phi(\hat{\theta}_n) - \phi(\theta) \right) \xrightarrow{d} (J\phi(\theta))X ,$$

and, more precisely,

$$r_n \left(\phi(\hat{\theta}_n) - \phi(\theta) \right) - r_n(J\phi(\theta))(\hat{\theta}_n - \theta) \xrightarrow{\mathbb{P}} 0 .$$

Remark. The second statement is the stronger one, since it says that the linearisation is exact up to a term negligible in probability at the scale r_n . Hence, the proof establishes the second statement and deduces the first from it.

Proof of the delta method

We denote by $R(h) = \phi(\theta + h) - \phi(\theta) - (J\phi(\theta))h$ the remainder of the first order expansion at θ , so that $R(0) = 0$ and $\|R(h)\| = o(\|h\|)$ as $h \rightarrow 0$, which is the definition of differentiability at θ .

Hence, we have

$$\begin{aligned} & r_n \left(\phi(\hat{\theta}_n) - \phi(\theta) \right) \\ &= (J\phi(\theta))r_n(\hat{\theta}_n - \theta) + r_n R(\hat{\theta}_n - \theta) \\ &= (J\phi(\theta))r_n(\hat{\theta}_n - \theta) + o_{\mathbb{P}} \left(\|r_n(\hat{\theta}_n - \theta)\| \right) && \text{by the transfer theorem with } q = 1 \\ &= (J\phi(\theta))r_n(\hat{\theta}_n - \theta) + o_{\mathbb{P}}(1) && \text{by Prohorov} \\ &\xrightarrow{d} (J\phi(\theta))X . && \text{by Slutsky and continuous mapping} \end{aligned}$$

Reading the hypotheses

Differentiability is needed at one point only. We assume nothing about ϕ away from θ , not even continuity, which is possible since $\hat{\theta}_n$ concentrates at θ .

The limit X need not be gaussian. The theorem holds for an arbitrary limit in law. However, in every application below, X is gaussian since it comes from the central limit theorem, and then $(J\phi(\theta))X \sim \mathcal{N}(0, J\phi(\theta)\Sigma(J\phi(\theta))^\top)$.

The rate is unchanged. Since the same r_n appears on both sides, $\phi(\hat{\theta}_n)$ converges at the rate of $\hat{\theta}_n$, with a different constant.

The case of a vanishing Jacobian. If $J\phi(\theta) = 0$, the limit is the constant 0 and the theorem, while true, only says that the rate r_n is too slow. We give the right rate in Part III.

Exercise, the inverse of a mean

Exercise. Let $X_1, X_2, \dots$ be i.i.d. of exponential law of parameter $\lambda > 0$, i.e. of density $x \mapsto \lambda e^{-\lambda x}$ on $(0, \infty)$, so that $\mathbb{E}_X[X] = 1/\lambda$ and $\text{Var}_X(X) = 1/\lambda^2$. Find the limit law of $\sqrt{n}(1/\bar{X}_n - \lambda)$.

Solution. By the central limit theorem, $\sqrt{n}(\bar{X}_n - 1/\lambda) \xrightarrow{d} \mathcal{N}(0, 1/\lambda^2)$. The map $\phi(x) = 1/x$ is differentiable at $1/\lambda$ with $\phi'(1/\lambda) = -\lambda^2$, so that we have

$$\begin{aligned}\sqrt{n} \left(\frac{1}{\bar{X}_n} - \lambda \right) &= \sqrt{n} (\phi(\bar{X}_n) - \phi(1/\lambda)) \\ &\xrightarrow{d} \mathcal{N}(0, \phi'(1/\lambda)^2 / \lambda^2) \quad \text{by the delta method with } r_n = \sqrt{n} \\ &= \mathcal{N}(0, \lambda^2) .\end{aligned}$$

Remark. The estimator is well defined since the X_i are positive almost surely. Its limiting variance depends on the unknown λ , which is the situation that variance stabilisation addresses in Part III.

Part II. Estimating a variance

The problem

Let $X_1, X_2, \dots$ be i.i.d. copies of a real X with $\mathbb{E}_X[|X|^4] < \infty$. We denote $\mu_1 = \mathbb{E}_X[X]$ and, for $\ell \in \{2, 3, 4\}$, the centered moments $\mu_\ell = \mathbb{E}_X[(X - \mu_1)^\ell]$.

We estimate the mean by $\hat{\mu}_{1,n} = \bar{X}_n$ and the variance μ_2 by the empirical variance

$$\hat{\mu}_{2,n} = \frac{1}{n} \sum_{i \leq n} (X_i - \bar{X}_n)^2 .$$

The difficulty. Since $\hat{\mu}_{1,n}$ is an average of i.i.d. variables, the central limit theorem applies to it directly. In contrast, $\hat{\mu}_{2,n}$ is not such an average, since its summands all involve $\bar{X}_n$, hence all the data, and they are identically distributed but not independent.

An identity for the empirical variance

We have

$$\begin{aligned}\hat{\mu}_{2,n} &= \frac{1}{n} \sum_{i \leq n} X_i^2 - \left(\frac{1}{n} \sum_{i \leq n} X_i \right)^2 && \text{by expanding the square} \\ &= \frac{1}{n} \sum_{i \leq n} (X_i - \mu_1)^2 - \left(\frac{1}{n} \sum_{i \leq n} (X_i - \mu_1) \right)^2 && \text{by invariance of a variance under translation}\end{aligned}$$

The right side is thus a fixed function of two averages of i.i.d. variables. Let us set $\phi(x, y) = y - x^2$ and define the vectors of $\mathbb{R}^2$

$$Y_i = \begin{pmatrix} X_i - \mu_1 \\ (X_i - \mu_1)^2 \end{pmatrix},$$

so that $\hat{\mu}_{2,n} = \phi(\bar{Y}_n)$ exactly, and $\mu_2 = \phi(0, \mu_2)$.

The central limit theorem for the pair

Since $\mathbb{E}_X[(X - \mu_1)] = 0$ and $\mathbb{E}_X[(X - \mu_1)^2] = \mu_2$, the vector Y_1 has mean $(0, \mu_2)^\top$ and covariance $\Sigma_Y := \text{Cov}_X(Y_1)$, given by

$$\Sigma_Y = \begin{pmatrix} \mu_2 & \mu_3 \\ \mu_3 & \mu_4 - \mu_2^2 \end{pmatrix},$$

which is finite since $\mathbb{E}_X[|X|^4] < \infty$. The entries are $\text{Var}_X(X - \mu_1) = \mu_2$, then $\mathbb{E}_X[(X - \mu_1)^3] = 0 \cdot \mu_2 = \mu_3$, and $\text{Var}_X((X - \mu_1)^2) = \mu_4 - \mu_2^2$.

Since the Y_i are i.i.d. and square integrable, the central limit theorem in $\mathbb{R}^2$ yields

$$\sqrt{n} (\bar{Y}_n - (0, \mu_2)^\top) \xrightarrow{d} \mathcal{N}(0, \Sigma_Y).$$

Applying the delta method

The map $\phi(x, y) = y - x^2$ is differentiable with $J\phi(x, y) = (-2x, 1)$, so that at the evaluation point, $J\phi(0, \mu_2) = (0, 1)$. Hence, we have

$$\begin{aligned}\sqrt{n}(\hat{\mu}_{2,n} - \mu_2) &= \sqrt{n}(\phi(\bar{Y}_n) - \phi(0, \mu_2)) \\ &\xrightarrow{d} \mathcal{N}(0, (0, 1)\Sigma_Y(0, 1)^\top) \quad , \quad \text{by the delta method with } r_n = \sqrt{n}\end{aligned}$$

and the quadratic form picks out the lower right entry:

$$\sqrt{n}(\hat{\mu}_{2,n} - \mu_2) \xrightarrow{d} \mathcal{N}(0, \mu_4 - \mu_2^2) \quad .$$

The first coordinate of Y has disappeared since $J\phi$ vanishes there. In other words, the fluctuation of $\bar{X}_n$ contributes nothing at first order to the fluctuation of the empirical variance.

Remarks on the result

The limiting variance is $\text{Var}_X((X - \mu_1)^2)$. It is zero if and only if $(X - \mu_1)^2$ is almost surely constant, i.e. when X takes only the two values $\mu_1 \pm \sqrt{\mu_2}$, and for every other law, the limit is nondegenerate.

In the gaussian case, $\mu_4 = 3\mu_2^2$, so the limit is $\mathcal{N}(0, 2\mu_2^2)$, and the relative precision $\sqrt{n}(\hat{\mu}_{2,n} - \mu_2)/\mu_2$ has limiting variance 2 whatever the scale.

The bias correction. With $s_n^2 = \frac{n}{n-1}\hat{\mu}_{2,n}$ the unbiased version, we have $\sqrt{n}(s_n^2 - \hat{\mu}_{2,n}) = \frac{\sqrt{n}}{n-1}\hat{\mu}_{2,n} \xrightarrow{\mathbb{P}} 0$, so that by asymptotic equivalence, s_n^2 has the same limit law. Hence, unbiasedness at fixed n has no asymptotic content here.

Exercise, the empirical standard deviation

Exercise. Under the assumptions of this part, and if $\mu_2 > 0$, find the limit law of $\sqrt{n}(\sqrt{\hat{\mu}_{2,n}} - \sqrt{\mu_2})$. Specialise it to the gaussian case.

Solution. The map $\phi(t) = \sqrt{t}$ is differentiable at $\mu_2 > 0$ with $\phi'(\mu_2) = 1/(2\sqrt{\mu_2})$. Hence, starting from the limit law of the empirical variance, we have

$$\begin{aligned}\sqrt{n} \left(\sqrt{\hat{\mu}_{2,n}} - \sqrt{\mu_2} \right) &= \sqrt{n} (\phi(\hat{\mu}_{2,n}) - \phi(\mu_2)) \\ &\xrightarrow{d} \mathcal{N} \left(0, \frac{\mu_4 - \mu_2^2}{4\mu_2} \right) . \quad \text{by the delta method with } r_n = \sqrt{n}\end{aligned}$$

In the gaussian case, $\mu_4 = 3\mu_2^2$, and the limit is $\mathcal{N}(0, \mu_2/2)$.

Remark. We applied the delta method twice, which is legitimate since its only input is a limit in law. Equivalently, we can apply it once to $(x, y) \mapsto \sqrt{y - x^2}$ at $(0, \mu_2)$, whose Jacobian there is $(0, 1/(2\sqrt{\mu_2}))$ by the chain rule.

Part III. Two refinements

The case of a vanishing Jacobian

If $J\phi(\theta) = 0$, the delta method yields $r_n(\phi(\hat{\theta}_n) - \phi(\theta)) \xrightarrow{d} 0$, a true but empty statement which only says that the rate r_n is too slow. We now show that the correct rate is r_n^2 .

Theorem (second order delta method). Let $\phi : \mathbb{R}^p \rightarrow \mathbb{R}$ be twice differentiable at θ with $\nabla\phi(\theta) = 0$, and let $r_n(\hat{\theta}_n - \theta) \xrightarrow{d} X$ with $r_n \rightarrow \infty$. Then

$$r_n^2 \left(\phi(\hat{\theta}_n) - \phi(\theta) \right) \xrightarrow{d} \frac{1}{2} X^\top (H\phi(\theta)) X .$$

The limit is no longer gaussian. Even for X gaussian, a quadratic form in a gaussian vector is a weighted sum of independent χ_1^2 variables, with weights the eigenvalues of $\frac{1}{2}\Sigma^{1/2}(H\phi(\theta))\Sigma^{1/2}$.

Proof of the second order version

Using $\nabla\phi(\theta) = 0$, we denote by $R(h) = \phi(\theta + h) - \phi(\theta) - \frac{1}{2}h^\top (H\phi(\theta))h$ the remainder of the second order expansion at θ , so that $R(0) = 0$ and $|R(h)| = o(\|h\|^2)$ as $h \rightarrow 0$, which is the Taylor expansion at order two of a function twice differentiable at θ .

Hence, denoting $Z_n = r_n(\hat{\theta}_n - \theta)$, we have

$$\begin{aligned} & r_n^2 \left(\phi(\hat{\theta}_n) - \phi(\theta) \right) \\ &= \frac{1}{2} Z_n^\top (H\phi(\theta)) Z_n + r_n^2 R(\hat{\theta}_n - \theta) \\ &= \frac{1}{2} Z_n^\top (H\phi(\theta)) Z_n + o_{\mathbb{P}}(\|Z_n\|^2) && \text{by the transfer theorem with } q = 2 \\ &= \frac{1}{2} Z_n^\top (H\phi(\theta)) Z_n + o_{\mathbb{P}}(1) && \text{by Prohorov} \\ &\xrightarrow{d} \frac{1}{2} X^\top (H\phi(\theta)) X . && \text{by Slutsky and continuous mapping} \end{aligned}$$

Example, a proportion at one half

Let $X_1, \dots, X_n$ be i.i.d. $\text{Ber}(\theta)$ and $\phi(\theta) = \theta(1 - \theta)$, the variance of a single observation. Then $\phi'(\theta) = 1 - 2\theta$, which vanishes only at $\theta = 1/2$.

Away from one half, the delta method yields $\sqrt{n}(\bar{X}_n(1 - \bar{X}_n) - \theta(1 - \theta)) \xrightarrow{d} \mathcal{N}(0, (1 - 2\theta)^2\theta(1 - \theta))$.

At $\theta = 1/2$, the first order limit is degenerate, and applying the second order version with $\phi'' = -2$ and $\sqrt{n}(\bar{X}_n - \frac{1}{2}) \xrightarrow{d} \mathcal{N}(0, \frac{1}{4})$, we obtain, with $Z \sim \mathcal{N}(0, \frac{1}{4})$,

$$\begin{aligned} n(\bar{X}_n(1 - \bar{X}_n) - 1/4) &\xrightarrow{d} -Z^2 \\ &\stackrel{d}{=} -\frac{1}{4}\chi_1^2. \end{aligned}$$

The rate improves from $n^{-1/2}$ to n^{-1} and the limit is one sided, as expected since $\bar{X}_n(1 - \bar{X}_n) \leq 1/4$ always, so that the error can only have one sign.

Exercise, a cosine at zero

Exercise. Let $X_1, X_2, \dots$ be i.i.d. real with $\mathbb{E}_X[X] = 0$ and $\text{Var}_X(X) = \sigma^2 \in (0, \infty)$. Find the limit law of $\sqrt{n}(\cos \bar{X}_n - 1)$, then the correct rate and the corresponding limit law.

Solution. By the central limit theorem, $\sqrt{n}\bar{X}_n \xrightarrow{d} \mathcal{N}(0, \sigma^2)$. Since $\cos'(0) = -\sin 0 = 0$, the delta method yields $\sqrt{n}(\cos \bar{X}_n - 1) \xrightarrow{d} 0$, so that the rate $n^{-1/2}$ is too slow.

Since $\cos''(0) = -1$, the second order version with $r_n = \sqrt{n}$ yields, with $Z \sim \mathcal{N}(0, \sigma^2)$,

$$\begin{aligned} n(\cos \bar{X}_n - 1) &\xrightarrow{d} -\frac{1}{2}Z^2 \\ &\stackrel{d}{=} -\frac{\sigma^2}{2}\chi_1^2. \end{aligned}$$

The limit is nonpositive, as it should be since $\cos \leq 1$.

Variance stabilisation

Let us take $p = m = 1$ and suppose that $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2(\theta))$, with a limiting variance depending on the unknown θ . For ϕ differentiable, the delta method yields

$$\sqrt{n} \left(\phi(\hat{\theta}_n) - \phi(\theta) \right) \xrightarrow{d} \mathcal{N} \left(0, \phi'(\theta)^2 \sigma^2(\theta) \right) .$$

The recipe. We choose ϕ with $\phi'(\theta) = 1/\sigma(\theta)$ for every θ , i.e. a primitive of $1/\sigma$. Then the limit is $\mathcal{N}(0, 1)$, the same for every θ , so that the unknown no longer appears in the limit law.

Advantages. Since no plug-in estimate of the variance is needed, the studentisation step of lecture 2 disappears. Furthermore, the normal approximation is usually accurate at smaller n , since the transformed statistic is closer to symmetric.

The arcsine transformation

For $X_1, \dots, X_n$ i.i.d. $\text{Ber}(\theta)$ with $\theta \in (0, 1)$, $\sigma^2(\theta) = \theta(1 - \theta)$, so that the recipe asks for a primitive of $1/\sqrt{\theta(1 - \theta)}$, namely

$$\phi(\theta) = 2 \arcsin \sqrt{\theta} ,$$

since $\phi'(\theta) = \frac{2}{\sqrt{1-\theta}} \cdot \frac{1}{2\sqrt{\theta}} = 1/\sqrt{\theta(1 - \theta)}$, hence $\sqrt{n}(2 \arcsin \sqrt{\bar{X}_n} - 2 \arcsin \sqrt{\theta}) \xrightarrow{d} \mathcal{N}(0, 1)$.

The interval. With r_α the quantile of order $1 - \alpha/2$ of $\mathcal{N}(0, 1)$, inverting ϕ , which is increasing from $[0, 1]$ onto $[0, \pi]$, yields an asymptotic confidence interval of level $1 - \alpha$ for θ , where the angles are clipped to $[0, \pi/2]$:

$$\left[\sin^2 \left(\max \left(0, \arcsin \sqrt{\bar{X}_n} - r_\alpha / (2\sqrt{n}) \right) \right), \sin^2 \left(\min \left(\pi/2, \arcsin \sqrt{\bar{X}_n} + r_\alpha / (2\sqrt{n}) \right) \right) \right] .$$

Another classical case. Poisson with $\sigma^2(\lambda) = \lambda$ leads to $\phi(\lambda) = 2\sqrt{\lambda}$.

The Fisher transformation

Let $(X_1, Y_1), (X_2, Y_2), \dots$ be i.i.d. gaussian pairs of $\mathbb{R}^2$ with positive variances and correlation $\rho \in (-1, 1)$. We estimate ρ by the empirical correlation coefficient

$$r_n = \frac{\sum_{i \leq n} (X_i - \bar{X}_n)(Y_i - \bar{Y}_n)}{(\sum_{i \leq n} (X_i - \bar{X}_n)^2 \sum_{i \leq n} (Y_i - \bar{Y}_n)^2)^{1/2}} .$$

Its limit law. Since r_n is a smooth function of empirical moments, the delta method yields, as we prove in the exercise on the correlation coefficient,

$$\sqrt{n}(r_n - \rho) \xrightarrow{d} \mathcal{N}(0, (1 - \rho^2)^2) .$$

The transformation. Here $\sigma(\rho) = 1 - \rho^2$, whose inverse has the primitive $\operatorname{arctanh}(\rho) = \frac{1}{2} \log \frac{1+\rho}{1-\rho}$ on $(-1, 1)$. Hence, $\sqrt{n}(\operatorname{arctanh}(r_n) - \operatorname{arctanh}(\rho)) \xrightarrow{d} \mathcal{N}(0, 1)$, and mapping an interval for $\operatorname{arctanh}(\rho)$ back by $\tanh$ yields an interval for ρ inside $(-1, 1)$.

The variance $(1 - \rho^2)^2$ relies on the fourth moments of the gaussian law. For other laws, the limiting variance of r_n is in general different, and the intervals built this way lose their nominal level.

Exercise, the correlation coefficient

Exercise. In the setting of the Fisher transformation, prove that $\sqrt{n}(r_n - \rho) \xrightarrow{d} \mathcal{N}(0, (1 - \rho^2)^2)$. We recall that, for a gaussian pair (X, Y) with centered coordinates of unit variance and correlation ρ , we have $\mathbb{E}_{X,Y}[X^4] = 3$, $\mathbb{E}_{X,Y}[X^3Y] = 3\rho$ and $\mathbb{E}_{X,Y}[X^2Y^2] = 1 + 2\rho^2$.

Reduction. Since r_n and ρ are invariant under increasing affine maps of each coordinate, we may assume that X_1 and Y_1 are centered with unit variance. Then $r_n = \phi(\bar{Z}_n)$, with $Z_i = (X_i, Y_i, X_i^2, Y_i^2, X_iY_i)^\top$ and $\phi(s, t, u, v, w) = (w - st) / \sqrt{(u - s^2)(v - t^2)}$.

Delta method. At $\mathbb{E}_{X,Y}[Z_1] = (0, 0, 1, 1, \rho)^\top$, the Jacobian of ϕ is $(0, 0, -\rho/2, -\rho/2, 1)$, the mean coordinates disappearing as for the empirical variance. Hence, the limit is centered gaussian with the variance of $W = X_1Y_1 - \rho(X_1^2 + Y_1^2)/2$, and expanding W^2 with $\mathbb{E}_{X,Y}[W] = 0$, we have

$$\begin{aligned}\text{Var}_{X,Y}(W) &= (1 + 2\rho^2) - 6\rho^2 + \rho^2(2 + \rho^2) \quad \text{by the gaussian moments} \\ &= (1 - \rho^2)^2.\end{aligned}$$

Exercise, stabilising an exponential rate

Exercise. For $\hat{\lambda}_n = 1/\bar{X}_n$, the exercise on the inverse of a mean yields $\sqrt{n}(\hat{\lambda}_n - \lambda) \xrightarrow{d} \mathcal{N}(0, \lambda^2)$. Find a variance stabilising transformation, and deduce an asymptotic confidence interval of level $1 - \alpha$ for λ .

Solution. Here $\sigma(\lambda) = \lambda$, and a primitive of $1/\lambda$ on $(0, \infty)$ is $\phi = \log$. Since $\phi'(\lambda)^2 \sigma^2(\lambda) = 1$, the delta method yields

$$\sqrt{n} \left(\log \hat{\lambda}_n - \log \lambda \right) \xrightarrow{d} \mathcal{N}(0, 1) .$$

With r_α the quantile of order $1 - \alpha/2$ of $\mathcal{N}(0, 1)$, the probability that $|\log \hat{\lambda}_n - \log \lambda| \leq r_\alpha / \sqrt{n}$ thus tends to $1 - \alpha$ by portmanteau, and exponentiating yields the interval

$$\left[\hat{\lambda}_n e^{-r_\alpha / \sqrt{n}}, \hat{\lambda}_n e^{r_\alpha / \sqrt{n}} \right] .$$

It lies in $(0, \infty)$ for every n , in contrast with the plug-in interval $\hat{\lambda}_n(1 \pm r_\alpha / \sqrt{n})$, whose lower end is negative when $n < r_\alpha^2$.

Part IV. The method of moments

The model

Definition. A parametric statistical model is a family $\{\mathcal{L}_\theta; \theta \in \Theta\}$ with $\Theta \subset \mathbb{R}^p$, each $\mathcal{L}_\theta$ a law on $\mathbb{R}^d$. In other words, it is a set of candidate laws for X_1 .

Well specified. We assume throughout that the model contains the truth, i.e. that there is θ_0 in the interior of Θ such that $X_1, \dots, X_n$ are i.i.d. of law $\mathcal{L}_{\theta_0}$, and our goal is to estimate θ_0 .

Notation. $\mathbb{E}_\theta, \mathbb{P}_\theta, \text{Cov}_\theta$ denote the expectation, probability and covariance computed as if the parameter were θ . For instance, with $d = 1, \Theta = (0, \infty)$ and $\mathcal{L}_\theta = \mathcal{N}(0, \theta)$, we have

$$\begin{aligned}\mathbb{E}_3[X_1^2] &= \int_{\mathbb{R}} x^2 d\mathcal{L}_3(x) \\ &= \mathbb{E}_{Z \sim \mathcal{N}(0,3)}[Z^2] \\ &= 3 .\end{aligned}$$

We keep writing $\mathbb{E}_{\theta_0} = \mathbb{E}_X$, since $\mathcal{L}_{\theta_0}$ really is the law of the data.

The method

Method of moments. We choose $f_1, \dots, f_p : \mathbb{R}^d \rightarrow \mathbb{R}$ and we look for a parameter $\hat{\theta}$ at which the theoretical moments equal the empirical ones:

$$\begin{cases} \frac{1}{n} \sum_{i \leq n} f_1(X_i) = \mathbb{E}_{\hat{\theta}}[f_1(X_1)] \\ \vdots \\ \frac{1}{n} \sum_{i \leq n} f_p(X_i) = \mathbb{E}_{\hat{\theta}}[f_p(X_1)] . \end{cases}$$

Compactly. With the **moment map** $e(\theta) = (\mathbb{E}_{\theta}[f_1(X_1)], \dots, \mathbb{E}_{\theta}[f_p(X_1)])^\top$ and the empirical moments $e_n = \frac{1}{n} \sum_{i \leq n} (f_1(X_i), \dots, f_p(X_i))^\top$, the system takes the form $e(\hat{\theta}) = e_n$, so that $\hat{\theta} = e^{-1}(e_n)$ whenever the inverse is available.

Intuition. The law of large numbers yields $e_n \xrightarrow{\mathbb{P}} e(\theta_0)$, and since e^{-1} is continuous near $e(\theta_0)$, the solution should be close to θ_0 . The analysis below makes this precise and adds the rate.

Exercise, the gaussian model

Exercise. Let $\Theta = \mathbb{R} \times (0, \infty)$, $\theta = (m, \sigma^2)$ and $\mathcal{L}_\theta = \mathcal{N}(m, \sigma^2)$. With $f_1(x) = x$ and $f_2(x) = x^2$, find the estimator.

Solution. We have $\mathbb{E}_\theta[f_1(X_1)] = m$ and $\mathbb{E}_\theta[f_2(X_1)] = m^2 + \sigma^2$, so that the system is

$$\begin{aligned}\frac{1}{n} \sum_{i \leq n} X_i &= \hat{m}_n, \\ \frac{1}{n} \sum_{i \leq n} X_i^2 &= \hat{m}_n^2 + \hat{\sigma}_n^2.\end{aligned}$$

It has the unique solution $\hat{m}_n = \bar{X}_n$ and $\hat{\sigma}_n^2 = \frac{1}{n} \sum_{i \leq n} X_i^2 - \bar{X}_n^2$, that is,

$$\hat{\sigma}_n^2 = \frac{1}{n} \sum_{i \leq n} (X_i - \bar{X}_n)^2,$$

i.e. the empirical mean and the empirical variance. Their joint asymptotic normality follows from Part II, and the theorem below recovers it from the general statement.

Identifiability

The method of moments matches p features of the law through the moment map e , and it can only work if these features separate the parameters.

Identifiability from the moments is the requirement that $e(\theta) \neq e(\theta_0)$ for every $\theta \neq \theta_0$. Without it, two parameters produce identical moment equations, and no sample size separates them.

The theorem below assumes instead that e is continuously differentiable near θ_0 with $Je(\theta_0)$ invertible. By the inverse function theorem, this only forces local injectivity, which does not imply identifiability, and makes e locally invertible, which is what the estimator relies on.

Number of moment functions. There are as many moment functions as parameters (p of each), since $Je(\theta_0)$ is $p \times p$ and has to be invertible. With fewer, the system is underdetermined, whereas with more, it is generally incompatible.

Exercise, a gaussian scale model

Exercise. Let $\Theta = \mathbb{R} \setminus \{0\}$ and $\mathcal{L}_\theta = \mathcal{N}(0, \theta^2)$, with $p = 1$. Compute e , and decide whether θ_0 is identifiable from the moments and whether $Je(\theta_0)$ is invertible, first with $f_1(x) = x$, then with $f_1(x) = x^2$.

With $f_1(x) = x$, we have $e(\theta) = \mathbb{E}_\theta[X_1] = 0$ for every θ . Hence, e is constant, so that θ_0 is not identifiable and $Je(\theta_0) = 0$ is not invertible.

With $f_1(x) = x^2$, we have $e(\theta) = \mathbb{E}_\theta[X_1^2] = \theta^2$, so that $Je(\theta_0) = 2\theta_0 \neq 0$ is invertible. However, $e(-\theta_0) = e(\theta_0)$, so that θ_0 is not identifiable, which is an instance of local injectivity without identifiability.

The cause is the model. Since $\mathcal{L}_{-\theta} = \mathcal{L}_\theta$, no method separates θ_0 from $-\theta_0$. With $\Theta = (0, \infty)$ instead, $f_1(x) = x^2$ identifies θ_0 , and the method yields $\hat{\theta}_n = (\frac{1}{n} \sum_{i \leq n} X_i^2)^{1/2}$.

Asymptotic normality

Theorem. With e as above, assume

1. θ_0 is in the interior of Θ and there is $\epsilon > 0$ with $B(\theta_0, \epsilon) \subset \Theta$,
2. e is continuously differentiable on $B(\theta_0, \epsilon)$ with $Je(\theta_0)$ invertible,
3. $\mathbb{E}_X[|f_j(X_1)|^2] < \infty$ for $j = 1, \dots, p$.

Then we can define a random vector $\hat{\theta}_n$ that solves the moment system with probability going to 1, and

$$\sqrt{n} \left(\hat{\theta}_n - \theta_0 \right) \xrightarrow{d} \mathcal{N} \left(0, (Je(\theta_0))^{-1} \Sigma_f (Je(\theta_0))^{-\top} \right) ,$$

where Σ_f is the covariance matrix of the random vector $(f_1(X_1), \dots, f_p(X_1))$.

Proof, local inversion

Since e is continuously differentiable near θ_0 with $Je(\theta_0)$ invertible, the inverse function theorem yields open sets $U \ni \theta_0$ and $V \ni e(\theta_0)$ such that $e : U \rightarrow V$ is a C^1 diffeomorphism, and, for every $v = e(u) \in V$,

$$(Je^{-1})(v) = (Je(u))^{-1} .$$

In particular, at the point of interest,

$$(Je^{-1})(e(\theta_0)) = (Je(\theta_0))^{-1} .$$

We only use hypotheses 1 and 2 here, and they allow us to apply the delta method to $\phi = e^{-1}$ at $e(\theta_0)$.

Proof, existence with probability going to one

By hypothesis 3, each $f_j(X_i)$ is square integrable, hence integrable on a probability space, so that the law of large numbers applies coordinatewise:

$$e_n \xrightarrow{\mathbb{P}} e(\theta_0) .$$

Since V is open and contains $e(\theta_0)$, this yields $\mathbb{P}(e_n \in V) \rightarrow 1$, and we define

$$\hat{\theta}_n = \begin{cases} e^{-1}(e_n) & \text{if } e_n \in V \\ \text{an arbitrary value} & \text{if } e_n \notin V . \end{cases}$$

On the event $\{e_n \in V\}$, we have $e(\hat{\theta}_n) = e_n$, so that $\hat{\theta}_n$ solves the moment system, and this event has probability going to 1. The estimator is to be understood in this sense, since the system may have no solution for small n , which does not matter asymptotically.

Proof, removing the arbitrary value

We define a companion sequence that always lies in V ,

$$\tilde{e}_n = \begin{cases} e_n & \text{if } e_n \in V \\ e(\theta_0) & \text{if } e_n \notin V, \end{cases}$$

so that $e^{-1}(\tilde{e}_n)$ is always defined, and equals $\hat{\theta}_n$ on $\{e_n \in V\}$.

Since $\theta_0 = e^{-1}(e(\theta_0))$, the two sequences $\sqrt{n}(\hat{\theta}_n - \theta_0)$ and $\sqrt{n}(e^{-1}(\tilde{e}_n) - e^{-1}(e(\theta_0)))$ coincide on $\{e_n \in V\}$, hence for every $\epsilon > 0$

$$\mathbb{P} \left(\left\| \sqrt{n}(\hat{\theta}_n - \theta_0) - \sqrt{n}(e^{-1}(\tilde{e}_n) - e^{-1}(e(\theta_0))) \right\| \geq \epsilon \right) \leq \mathbb{P}(e_n \notin V),$$

which goes to 0.

By asymptotic equivalence, it is thus enough to find the limit law of $\sqrt{n}(e^{-1}(\tilde{e}_n) - e^{-1}(e(\theta_0)))$.

Proof, the delta method

Since the vectors $(f_1(X_i), \dots, f_p(X_i)), i \leq n$, are i.i.d., square integrable by hypothesis 3, with mean $e(\theta_0)$ and covariance Σ_f , the central limit theorem yields

$$\sqrt{n}(e_n - e(\theta_0)) \xrightarrow{d} \mathcal{N}(0, \Sigma_f) ,$$

and since $\tilde{e}_n$ differs from e_n only on an event of vanishing probability, asymptotic equivalence yields the same limit for $\sqrt{n}(\tilde{e}_n - e(\theta_0))$.

We now apply the delta method to $\phi = e^{-1}$, differentiable at $e(\theta_0)$, with $r_n = \sqrt{n}$, and we obtain

$$\begin{aligned} \sqrt{n}(e^{-1}(\tilde{e}_n) - e^{-1}(e(\theta_0))) &\xrightarrow{d} (Je(\theta_0))^{-1} \mathcal{N}(0, \Sigma_f) && \text{by local inversion} \\ &= \mathcal{N}(0, (Je(\theta_0))^{-1} \Sigma_f (Je(\theta_0))^{-\top}) . \end{aligned}$$

Together with the removal of the arbitrary value, this concludes the proof.

Exercise, the gaussian model, continued

Exercise. In the gaussian model with $f_1(x) = x$, $f_2(x) = x^2$ and $\theta_0 = (m_0, \sigma_0^2)$, compute the asymptotic covariance of $\sqrt{n}((\hat{m}_n, \hat{\sigma}_n^2) - (m_0, \sigma_0^2))$ given by the theorem.

Solution. Since $e(m, \sigma^2) = (m, m^2 + \sigma^2)^\top$, the inverse of $Je(\theta_0)$ has rows $(1, 0)$ and $(-2m_0, 1)$. Using $\mathbb{E}_X[X_1^3] = m_0^3 + 3m_0\sigma_0^2$ and $\mathbb{E}_X[X_1^4] = m_0^4 + 6m_0^2\sigma_0^2 + 3\sigma_0^4$, we obtain

$$\Sigma_f = \begin{pmatrix} \sigma_0^2 & 2m_0\sigma_0^2 \\ 2m_0\sigma_0^2 & 4m_0^2\sigma_0^2 + 2\sigma_0^4 \end{pmatrix}.$$

Hence, we have

$$\begin{aligned} (Je(\theta_0))^{-1} \Sigma_f (Je(\theta_0))^{-\top} &= \begin{pmatrix} \sigma_0^2 & 2m_0\sigma_0^2 \\ 0 & 2\sigma_0^4 \end{pmatrix} \begin{pmatrix} 1 & -2m_0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} \sigma_0^2 & 0 \\ 0 & 2\sigma_0^4 \end{pmatrix}. \end{aligned}$$

We recover the limit $\mathcal{N}(0, 2\sigma_0^4)$ of Part II for the empirical variance, and the two estimators are asymptotically independent, since the limit is a gaussian vector with diagonal covariance.

Reading the asymptotic variance

Let us take $p = 1$, so that, with $e(\theta) = \mathbb{E}_\theta[f_1(X_1)]$, the asymptotic variance is the number

$$\frac{\text{Var}_X(f_1(X_1))}{(e'(\theta_0))^2}.$$

The numerator is the noise. The smaller the fluctuation of $f_1(X_1)$, the more precisely the empirical moment estimates the theoretical one.

The denominator is the signal. The larger $|e'(\theta_0)|$, the more the theoretical moment moves when the parameter moves, and thus the better f_1 discriminates between θ_0 and its neighbours.

A good moment function thus has an expectation that varies fast with θ and a small variance under $\mathcal{L}_{\theta_0}$. However, the method does not prescribe f_1 , and different choices lead to different variances, none of them optimal in general.

Exercise, two moment functions

Exercise. Let $\Theta = (0, \infty)$ and $\mathcal{L}_\theta$ the exponential law of parameter θ , for which $\mathbb{E}_\theta[X_1^k] = k!/\theta^k$.

Compute the asymptotic variance of the method of moments with $f_1(x) = x$, then with $f_1(x) = x^2$, and compare.

With $f_1(x) = x$, we have $e(\theta) = 1/\theta$, so that $e'(\theta_0)^2 = 1/\theta_0^4$, and $\text{Var}_X(X_1) = 1/\theta_0^2$. The asymptotic variance is thus θ_0^2 , as in the exercise on the inverse of a mean, since $\hat{\theta}_n = 1/\bar{X}_n$.

With $f_1(x) = x^2$, we have $e(\theta) = 2/\theta^2$, so that $e'(\theta_0)^2 = 16/\theta_0^6$, and $\text{Var}_X(X_1^2) = (24 - 4)/\theta_0^4$. Hence, the asymptotic variance is

$$\frac{20/\theta_0^4}{16/\theta_0^6} = \frac{5}{4}\theta_0^2 .$$

Comparison. The first moment is better for every θ_0 , by a factor $5/4$. Indeed, moving from x to x^2 multiplies the standard deviation of the noise by $\sqrt{20}/\theta_0$, but the signal $|e'(\theta_0)|$ only by $4/\theta_0$.

The limits of the method

Example. Let $\Theta = (0, \infty)$, $\mathcal{L}_\theta = \text{Unif}([0, \theta])$ and $f_1(x) = x$. Then $e(\theta) = \theta/2$, so that $e'(\theta_0) = 1/2$ is invertible, $\Sigma_f = \text{Var}_X(X) = \theta_0^2/12$, and the theorem applies to $\hat{\theta}_n = 2\bar{X}_n$:

$$\sqrt{n} \left(\hat{\theta}_n - \theta_0 \right) \xrightarrow{d} \mathcal{N} \left(0, \frac{\theta_0^2}{3} \right) .$$

Comparison with the maximum of the sample. For $M_n = \max_{i \leq n} X_i$, a direct computation yields $\mathbb{P}(n(\theta_0 - M_n) > t) = (1 - t/(n\theta_0))^n \rightarrow e^{-t/\theta_0}$, so that $n(\theta_0 - M_n)$ converges in law to an exponential. Hence, the rate is $1/n$ rather than $1/\sqrt{n}$.

Interpretation. The method of moments is consistent and asymptotically normal here, but badly suboptimal, since the information about θ_0 lies in the largest observation rather than in an average. No choice of moment functions repairs this.